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DESIGN MANUAL	1

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Overview

Introduction

Sunswift Racing is a unique and highly complex team, working on more than one hundred projects that bring together multiple engineering disciplines including mechanical, electrical, software, and chemical, into a single, cohesive system. However, integrating these diverse fields presents challenges in clear communication and standardisation throughout the design process. Each discipline has its own technical language, tools, and methodologies, and without a structured approach, misalignment can lead to inefficiencies, design inconsistencies, and gaps in knowledge transfer. As Sunswift evolves with each iteration of the car, the team must also adapt. To support this continuous improvement, the Systems Engineering team has developed an updated version of the Solution Guide (The Design Manual), designed to work alongside newly created discipline-specific guides for a more consistent and structured approach to design.

Role of the Design Manual

While the Solution Guide is highly detailed and comprehensive, its complexity and length make it difficult for engineers to quickly identify key requirements at each stage, track additions from one stage to the next, and understand how designs should evolve.
The distinctions between stages are not always clear, making navigation more challenging. Additionally, some disciplines lack sufficient guidance, reducing the document's practicality and hindering a consistent design approach across the broader project scope.

To address these challenges, the **Design Manual** was created as a foundational guide for all Sunswift engineers, providing an overarching, structured framework that ensures consistency across all design stages. It includes general, concise checklists applicable to all disciplines, ensuring alignment to team targets, before engineers refer to their specific discipline guides for in-depth technical details. This manual helps engineers understand core requirements, goals, and expectations, ultimately improving their design process and aligning with Sunswift's objectives of innovation and high-quality engineering.

The Design Manual bridges existing gaps by:

- Standardising documentation to enhance knowledge retention, streamline onboarding, and ensure design consistency across project iterations, while providing discipline-specific guidance in a consistent format."

- Facilitating cross-disciplinary collaboration by improving communication and alignment between different engineering teams.
- Streamlining verification and validation (V&V) processes to meet ADR and other regulatory requirements.

The Design Manual ensures that every ream member has a clear, structured roadmap, strengthening consistency and collaboration, and enhancing the project success across all engineering disciplines.

Discipline Guides

Every design engineer needs to refer to **two key documents** during the design process:

1. The Design Manual
2. Discipline Guides: These contain discipline-specific requirements, methodologies, and best practices, used to complement the Design Manual by offering detailed technical guidance tailored to each field. The discipline guides consist of: Mechanical, Electrical, Chemical and Software.

Documentation

Sunswift Racing implements consistent and structured documentation for maintaining knowledge, ensuring uniformity, and tracking design progress. Follow these steps to keep records organised and accessible:

1. Create a word document if not existing already on SharePoint under “Technical/Sunswift 8/Project Name”. See this template <link>.
2. Save it as “SRP08-XXX-Design Document”.
3. Follow the design phases outlined in this manual, documenting each phase as a **new document.**
4. Once a phase is complete, have it reviewed by your systems liaison. The review process for each stage is detailed below.

The Design Process

The Sunswift design process follows six sequential phases (Conceptual, Preliminary, Detailed, Manufacturing, Assembly & Testing and Retirement), each building on the last with increasing depth of development and technical detail, guiding projects from concept to final implementation. Engineers should:

- Understand the objectives and expectations of each phase.
- Follow the phase-specific requirements checklist.
- Document progress and coordinate a design review with systems liaisons.
- Upon approval, start the next phase in a new document.

The use of this Design Manual alongside the Discipline Guides ensures consistency, alignment with Sunswift's engineering standards, and a streamlined workflow for all design engineers.

Requirements

Requirements are statements that constrain the design of the vehicle. They are structured by type, where different types are addressed in specific phases. Systems engineers will be part of the process to guide formulation. The formulation of requirements should be at the start of each design phase. Specifically, the requirements should be before exploring and digging into design concepts.

Note: Please contact any system engineers before start working on it. Requirements should be expanded on from conceptual to preliminary and then addressed in detailed design. Each requirement shall be numbered based on the checklist item it belongs to (checklist items for structure of each design report provided below). For example, the 2nd requirement in 2.2.1 Teamwide specifications should be referred to as 2.2.1.2 and follow the below best practices.

Requirement Best Practices:

1. **Clear:** Minimise usage of “and”, ambiguous words such as “user-friendly", “efficient”, “maximise”, etc. It should be a simple specific sentence.
2. **Necessary:** If nothing was impacted if the requirement was not included, then it is deemed unnecessary.
3. **Verifiable:** Establish a qualitative/quantitative criteria for which the design/performance of the project can be compared against.
    a. Include allowable tolerances due to unexpected surprises during the design and build process
4. **Attainable:** Requirements must be feasible within current constraints of the project.
5. **Independent:** Requirements must not contradict any existing ones.
6. **Regularly Refine:** communicated with the team, people to check over the requirements, make sure any changes are documented with reasons.

Conceptual Design	Preliminary Design	Detailed Design & Development
Functional Requirements: - Normal Operations **Performance Requirements**: - Teamwide Specifications **Interface Requirements:** - Project/System Interfaces **External Requirements** - ADRs - ISOs - UNECEs	**Conceptual Design** **Functional Requirements:** - Abnormal Operation: Emergency Scenarios - Abnormal Operation: User Error - Testing/Maintenance/Fault Diagnostic **Performance Requirements**: - Teamwide Specifications - Technical Specifications - Safety Specifications **Interface Requirements:** - Project/System Interfaces - Mounting Points/Areas/Routing **Reliability Requirements** **External Requirements** - ADRs - ISOs - UNECEs	**Preliminary Design** **Functional Requirements:** - Abnormal Operation: Emergency Scenarios - Abnormal Operation: User Error - Testing/Maintenance/Fault Diagnostic **Performance Requirements**: - Technical Specifications - Safety Specifications **Interface Requirements:** - Project/System Interface - Mounting Points/Areas/Routing - Dimensions Constraints **Reliability Requirements** **External Requirements** - ADRs - ISOs - UNECEs
Key Outcomes	**Key Outcomes**	**Key Outcomes**
Project definition Multiple Designs	Key Component's & Skeleton BOM Requirements & Interfaces	Design Dimensions & Components Technical Documentation

Conceptual Design

The conceptual design phase establishes the project's high-level scope and explores multiple potential design solutions. The focus is on defining the project's purpose, function, and role within the Sunswift's overall vehicle system, while also considering feasibility and key requirements.

At this stage, project leads should conduct comprehensive research and generate several design concepts, assess their viability, and compare them based on relevant design factors. The final selected concept will progress to the Preliminary Design phase, where it will be refined and explored in greater detail.

To get a conceptual design approved, the Conceptual Design Review Process is outlined here <link>

	CONCEPTUAL DESIGN CHECKLIST	
	Deliverable	**Criteria**
1	**Project**	
1.1	Project Scope	Discuss the high-level function of the project, and outline the components and systems involved, defining the project purpose and role within the overall design.
1.2	Assigned Departments	
1.2.1	Primary Department	Identify the main department responsible for the project, that will oversee the development and delivery of the design.
1.2.2	Supporting Departments	Outline secondary departments that will support the project and provide assistance or expertise.
1.2.3	Team Members	List key team members, their roles and responsibilities within the project.
1.3	Project Interfaces	Identify and document all project interfaces with other systems or components that may impact functionality, compatibility, or integration.
2	**Requirements**	
2.1	Functional Requirements	Define the key operational functions of the project.
2.1.1	Normal Operation	Outline the system's expected function and operation under normal conditions.
2.2	Performance Requirements	Establish initial performance expectations and key benchmarks for the project.
2.2.1	Teamwide Specifications	Define high-level performance objectives that align with team standards and project scope.
2.3	Interface Requirements	Define the fundamental interfacing requirements needed for the project to integrate with other systems or components.
2.3.1	Project/System Interfaces	Outline the initial requirements for how the project must interact with other systems, ensuring compatibility, seamless communication.

2.4	External Requirements	Identify any external regulations, industry standards, or compliance requirements that the project must adhere to, such as Australian Design Rules (ADRs) or other relevant guidelines. The ADR Database is a useful tool for this
3	**Design Factors**	
3.0	Concept Generation	Brainstorm and develop multiple design concepts for the project. Briefly evaluate advantages, disadvantages, feasibility, and alignment with project goals for each design (in a table). Then, choose 2 - 3 concepts to analyse in stages 3.1 - 3.6. It is important to consider if the components are being manufactured or procured.
3.1	Lifecycle Analysis	Provide an initial assessment of the concept lifecycle, considering material selection, sustainability, recyclability, and maintenance. Do this by first consulting with Material Science regarding any aspects mentioned above and including their thoughts on each concept.
3.2	Risk Analysis	Conduct an initial risk assessment, identifying technical, operational, and financial risks, while providing potential mitigation approaches or responses.
3.3	Innovation Analysis	Assess the level of innovation for each concept classifying as high, medium or low with justifications. Compare designs and highlight key innovative aspects, while considering cost and learning involved in each.
3.4	Design Timeline	
3.4.1	Manufacture and Procurement Plan	Identify whether the project will be produced in-house or sourced externally and highlight potential suppliers or industry partners involved.
3.4.2	Design and Production Plan	Provide an estimated timeline for each design stage and the overall duration of the project.
3.5	Educational Development	
3.5.1	Learning Outcomes	Identify student learning opportunities, particularly in technical understanding, problem-solving, engineering & design processes and industry exposure.
3.5.2	Upskilling	Highlight any upskilling opportunities that will arise from working on the project, improving practical skills and hands on expertise. Identify potential challenges and the need for training in specific areas.
3.6	Cost Analysis	Provide a rough estimate of the costs associated with each concept, classifying components as low, medium, or high cost. Briefly explore potential methods to reduce expenses.
3.7	Concept Selection	Evaluate and justify the final design choice based on key design factors, clearly explaining why it is the best option to proceed to the Preliminary Design phase.

Preliminary Design

The preliminary design phase builds upon the conceptual design by refining the chosen concept, defining specific requirements, and further developing system interfaces. This phase involves deeper technical analysis, ensuring all design elements meet functional, performance, safety and interfacing requirements while maintaining compatibility with existing systems. It takes a comprehensive approach to design factors, providing additional details and identifying initial timelines. By the end of this stage, the design should be well-defined, with clear parameters for detailed development. Individual project components or sections of code should be addressed, and documentation may integrate technical specifications, diagrams, CAD models, process flows or pseudocode.

To get a preliminary design approved, the Preliminary Design Review Process is outlined here <link>

	PRELIMINARY DESIGN CHECKLIST	
	Deliverable	**Criteria**
1	**Project**	
1.1	Assemblies	Define all key components / assemblies of the system, identifying what each part does and its role in the project.
1.2	Component Review	Identify which components can be manufactured together (same manufacturing plan --> same manufacturer and approximately same time, also see section 3.4.1)
2	**Requirements (Include specific values / targets to meet)**	
2.1	Functional Requirements	Expand on the functional requirements from the conceptual phase, incorporating additional details and refinements.
2.1.1	Normal Operation	Describe component-specific expectations and highlight any refinements based on further analysis.
2.1.2	Abnormal Operation	Describe how the system will behave and the operation of components under abnormal or unexpected conditions.
2.1.2.1	User Error	Explain the expected system response to incorrect inputs or misuse, including tolerance levels for improper operation.
2.1.2.2	Emergency Scenarios	Explain expected system behaviour in failure events or hazardous situations, detailing how components should react to part failures or environmental threats.
2.1.3	Testing, Maintenance & Fault Diagnostics	Describe functionalities designed for troubleshooting, maintenance, and verification, ensuring ease of system checks, repairs, and performance assessments.
2.2	Performance Requirements	Expand on performance expectations, providing specific values that are measurable where possible.

2.2.1	Teamwide Specifications	Refine system-wide performance benchmarks, ensuring alignment with project objectives and team goals.
2.2.2	Technical Specifications	Provide preliminary technical requirements for the project and each component, considering parameters e.g. size, weight, power consumption, data rates, and performance constraints.
2.2.3	Safety Specifications	Outline safety requirements that the project and each component must adhere to, ensuring the overall system meets necessary safety standards.
2.3	Interface Requirements	Expand on interface requirements, ensuring all changes are documented and communicated to relevant team members
2.3.1	Project/System Interfaces	Establish clear points of interaction between components or subsystems, providing initial evidence of integration and communication between parts.
2.3.2	Mounting Points / Areas / Routing	Specify the mounting and routing requirements for the project, ensuring they align with other system elements for secure and functional integration.
2.4	External Requirements	Check which ADRs are relevant to your project on the master sheet of the ADR Database then go to that checklist's sheet. If the requirement has been correctly assigned, change the 'right project assigned' column from 'pending' to 'yes'. Specify if you need to use an alternative method of testing instead of ADR (should only be needed for destructive tests). Your design will not be approved without all ADR requirements from the ADR database added and subsequently addressed in the next phase (detailed design phase). Please contact your systems liaison if unsure.
2.5	Reliability Requirements	Assess the reliability of each component, ensuring that it can withstand long-term use without frequent failures or significant degradation.
3	**Design Factors**	
3.1	Lifecycle Analysis	Assess the lifespan of each component, refining material, manufacturing, and maintenance considerations. Compare analysis to the conceptual phase, justifying any changes and identifying barriers to recyclability or sustainability. Propose strategies to minimise material usage, reduce environmental impacts, and enhance recycling or reuse. Do this by consulting with Material Science and including their thoughts regarding any aspects mentioned above.
3.2	Risk Analysis	Conduct a detailed risk analysis for each component, comparing findings to the conceptual phase. Evaluate potential failures, project delays, and unknowns, proposing mitigation strategies. Discuss encountered risks and any implemented or possible solutions.

3.3	Innovation Analysis	Evaluate innovation for each component using an innovation matrix. Justify rankings based on novelty or improvements over existing technologies, along with possible risks and challenges. Identify any changes from the conceptual phase.
3.4	Design Timeline	
3.4.1	Manufacture & Procurement Plan	Refine the manufacturing and procurement strategy, identifying suppliers, partnerships, and determining lead times. Justify sourcing decisions and outline key resources required for production.
3.4.2	Design & Production Plan	Develop a draft timeline and plan for the design and production process, defining major milestones, dependencies, and high-risk components requiring early procurement.
3.5	Educational Development	
3.5.1	Learning Outcomes	Evaluate the knowledge and skills already gained and expected to be acquired through the design process. This includes technical understanding, industry interactions, problem-solving experiences and any future educational outcomes.
3.5.2	Upskilling	Outline challenges encountered or upcoming and the skills needed to overcome them. Explain opportunities to learn new tools or techniques, such as CAD modelling, machining processes, advanced circuit design, chemical analysis techniques or software applications. Discuss efforts to improve workflow efficiency and refine existing processes.
3.6	Cost Analysis	Estimate the cost (provide value) of each component, including manufacturing, procurement, and any unknowns. Costs should be categorised as actual or forecasted, with actual costs supported by supplier quotations where possible. Identify any uncertainties and provide justifications for high costs or any unknown values. Describe cost reduction strategies implemented or potential opportunities for further reducing expenses.

Detailed Design

The detailed design phase finalises all project specifications, ensuring a fully developed design that meets all requirements, building upon previous stages by defining exact parameters for each component and providing supporting evidence and data. By the end of this phase, the entire design should be ready for manufacturing and testing, backed up by engineering analyses, simulations, and validation reports. Documentation may include CAD models, technical specifications, electrical schematics, code simulations, chemical process flow diagrams, cost analysis and detailed procurement / production plans.

To get a detailed design approved, the Detailed Design Review Process is outlined here <link>

	DETAILED DESIGN AND DEVELOPMENT CHECKLIST	
	Deliverable	**Criteria**
1	**Project**	
1.1	Assemblies	Define all components / assemblies of the system. Each requirement section of the detailed design checklist should apply to assemblies.
2	**Requirements (Address how requirements are satisfied for assemblies)**	
2.1	Functional Requirements	Document any refinements to preliminary requirements and demonstrate how the final design meets the requirements.
2.1.1	Normal Operation	Demonstrate how the design ensures normal operation, backed by evidence such as simulations, calculations, or testing plans.
2.1.2	Abnormal Operation	Demonstrate how the system can handle unexpected conditions and external disruptions.
2.1.2.2	User Error	Demonstrate that the system can handle improper inputs or incorrect operation, specify tolerances and addressing all possible cases.
2.1.2.1	Emergency Scenarios	Demonstrate how failure risks are mitigated through redundancy features, fail-safe mechanisms, or failure analysis.
2.1.3	Testing, Maintenance & Fault Diagnostics	Demonstrate how the design supports ease of maintenance and fault detection, highlighting any new features that enhance troubleshooting and repair processes. Provide specific testing and diagnostic methods to ensure long-term reliability.
2.2	Performance Requirements	Demonstrate how the final design meets all defined performance criteria.
2.2.1	Teamwide Specifications	Validate that the system meets the defined performance benchmarks, providing supporting evidence from simulations, testing, or analysis.

2.2.2	Technical Specifications	Confirm and finalise all technical specifications, ensuring compliance through analysis, calculations, or prototype testing.
2.2.3	Safety Specifications	Verify adherence to safety requirements, demonstrating compliance with necessary safety standards through documented validation methods.
2.3	Interface Requirements	Demonstrate how the final design will meet interfacing requirements, providing evidence such as calculations, simulations, or prototype testing.
2.3.1	Project/System Interfaces	Validate that all interfaces will function as intended, ensuring compatibility through documented analysis, simulations, or testing.
2.3.2	Mounting points / Areas / Routing	Validate the mounting and routing requirements are met, providing final specifications and ensuring compliance with structural, spatial, and mechanical constraints.
2.4	External Requirements	Demonstrate how all components will comply with all applicable external regulations providing validation through design reviews, simulations, or documentation. Do this by copy pasting the External Requirements section from your preliminary design report and adding evidence below each requirement. Identify the degree of verification method required with a detailed description of how it is done.
2.5	Reliability Requirements	Demonstrate the reliability of each component, providing evidence of ability to withstand long-term use without frequent failures or significant degradation.
3	**Design Factors (for each key component)**	
3.0		
3.1	Lifecycle Analysis	Finalise the lifecycle analysis with specific data on component lifespans, expected operational duration, and replacement schedules. Provide a detailed end-of-life plan, including disassembly, replacement processes, responsible parties, and estimated reassembly time. Justify material choices based on research, linking them to sustainability goals, and evaluate for feasibility and environmental impact. This should only be possible with the approval of Material Science, so include their insights regarding any aspects mentioned above.
3.2	Risk Analysis	Finalise the risk analysis for each component, categorising failure risks by severity and refining mitigation strategies. Update project delays and unknowns, ensuring all response strategies are well-defined and supported by relevant data.
3.3	Innovation Analysis	Refine the innovation analysis for each component, providing clear justification for rankings and addressing shortcomings or areas with limited innovation. Assess the risks associated with innovation, compare them to prior evaluations.
3.4	Design Timeline	
3.4.1	Manufacture & Procurement Plan	Finalise the procurement and manufacturing plan, specifying confirmed suppliers, partnerships, and production strategies

		for each component. Define responsibilities within the team and outline contingency measures for delays or sourcing issues. Provide a document that highlights how a part will be manufactured or how it may be procured.
3.4.2	Design & Production Plan	Establish a final production timeline with defined start and end dates for each component. Ensure alignment with related projects, validate dependencies, and confirm milestone targets.
3.5	Educational Development	
3.5.1	Learning Outcomes	Reflect on the entire design process and assess the impact of project involvement. Summarise key takeaways in technical knowledge, industry exposure, and personal growth.
3.5.2	Upskilling	Reflect on the technical and practical skills developed, along with any further upskilling opportunities. Evaluate alternative software, methods, or supplier relationships that could enhance efficiency. Discuss how the experience has contributed to skill development and future career readiness.
3.6	Cost Analysis	Provide a detailed cost analysis outlining all project expenses. Specify manufacturing or procurement costs for each component and compare them to the allocated budget, identifying any overruns. If costs exceed the budget, suggest reduction strategies such as design modifications, alternative materials, or supplier negotiations.
4	**Supporting Documentation**	
4.1	Engineering Evidence	Provide comprehensive technical documentation, including calculations, analyses, and justifications to demonstrate the feasibility of the design and validate key decisions.
4.1.1	Technical Calculations	Include relevant mathematical models, equations, simulations, and analytical methods used to justify design choices.
4.1.2	Proof of Work	Document the progress and completion of design tasks through logs, testing and validation reports, design review documentation, code repositories.
4.1.3	Project Diagrams	Provide system schematics, flowcharts, block diagrams, wiring diagrams, engineering drawings, CAD models, or other necessary visual representations.
4.2	Physical Prototypes	If applicable, develop and test prototypes to assess functionality, performance, and manufacturability.
4.3	Bill of Materials	List all required components and materials, including specifications, suppliers, and estimated costs. Follow the provided template (link).

Manufacturing

* Manufacturing checklist will be added shortly to outline key considerations during manufacturing. To proceed to manufacturing, the detailed design checklist including supporting documentation and detailed design review must be completed. *

While manufacturing is a critical part of the overall design process, the details of the manufacturing stage will be included in each discipline-specific guide rather than within the main Design Manual. This is because manufacturing, assembly, and testing processes vary significantly between disciplines due to design factors, interfacing and ADRs. What is relevant for mechanical components may differ entirely from what is appropriate for electrical or chemical systems. Embedding manufacturing expectations directly into each discipline's guide allows for clearer, more relevant instructions tailored to the specific needs and constraints of that discipline.

However, a checklist will still exist in the Design Manual to acknowledge the importance of manufacturing as a team-wide process, acknowledge key requirements and to direct project leads to the appropriate discipline-specific documentation. This structure promotes both consistency in documentation and flexibility across technical areas.